

WeldNet: A voxel-based deep learning network for point cloud annular weld seam detection

WANG Hui^{1,2}, RONG YouMin^{1,2}, XU JiaJun^{1,2}, XIANG SongMing^{1,2}, PENG YiFan^{1,2} & HUANG Yu^{1,2*}

¹ State Key Laboratory of Intelligent Manufacturing Equipment and Technology, Huazhong University of Science and Technology, Wuhan 430074, China;

² School of Mechanical Science and Engineering, Huazhong University of Science and Technology, Wuhan 430074, China

Received August 31, 2023; accepted December 5, 2023; published online March 27, 2024

Weld seam detection is an important part of automated welding. At present, few studies have been conducted on annular weld seams, and a lot of defects exist in the point cloud model of the tube sheet obtained by RGB-D cameras and photography methods. Aiming at the above problems, this paper proposed an annular weld seam detection network named WeldNet where a voxel feature encoding layer was adaptively improved for annular weld seams, the sparse convolutional network and region proposal network (RPN) were used to detect annular weld seam position, and an annular weld seam detection loss function was designed. Further, an annular weld seam dataset was established to train the network. Compared with the random sampling consistency (RANSAC) method, WeldNet has a higher detection accuracy, as well as a higher detection success rate which has increased by 23%. Compared with U-Net, WeldNet has been proven to achieve a better detection result, and the intersection over the union of the weld seam detection is improved by 17.8%.

deep learning, point cloud, weld seam detection, welding, annular weld seam

Citation: Wang H, Rong Y M, Xu J J, et al. WeldNet: A voxel-based deep learning network for point cloud annular weld seam detection. *Sci China Tech Sci*, 2024, 67: 1215–1225, <https://doi.org/10.1007/s11431-023-2569-1>

1 Introduction

Welding plays a significant role in the industry and has been recognized as an important processing method for the field. However, traditional welding mainly relies on manual operations. The development of society and the advancement of technology urgently drive the constant improvement in the level of welding automation. Currently, the application of welding robots has been increasing [1]. The existing welding robots are mainly based on the “teaching-playback” method [2], which cannot achieve fully automated welding. To realize automatic welding, automatically detecting weld seams is of great necessity and should be achieved. Plenty of work

on weld inspection has been studied. Li et al. [3] used a monocular camera to design a welding seam recognition method for the inspection robot, and the error of this method was less than ± 20 mm. Vithanage et al. [4] designed a novel high-temperature ultrasound phased array roller probe to perform robotically delivered in-process inspection of multi-welds, which was able to be operated at temperatures up to 350°C. Yang et al. [5] proposed a novel weld seam extraction system including a fringe projection system that consisted of two cameras and a DLP projector, as well as a three-dimensional (3D) weld seam extraction algorithm. Fan et al. [6,7] designed a welding seam tracking system to extract the starting point and feature points of the welding seam. In these methods, the information on the weld seam was mainly stored in an image or point cloud. Furthermore, Ma et al.

*Corresponding author (email: yuhuang7208@163.com)

[8,9] classified and extracted features of complex welds with deep learning methods. Although the acquisition and processing of images are more mature, the point cloud has two advantages over images. On the one hand, 3D information can be directly obtained, and on the other hand, the mathematical expression is more concise. In light of the advantages, research on the weld point cloud is much more important and promising.

The point cloud is a collection of points in a coordinate system, and the points are generally composed of 3D coordinate values. It is an important way to digitally represent the 3D world with a wide range of applications, e.g., autonomous driving [10], agriculture [11], archaeology [12], medical care [13], welding, etc. In the field of welding, although there are a lot of methods for obtaining the point cloud, the accuracy of the obtained point clouds varies greatly. Gao et al. [14] obtained the point cloud of the D-type weldment through a 3D scanner whose accuracy reached ± 0.02 mm. Geng et al. [15] used a 3D surface scanning industrial camera to obtain the point cloud model of a steel mesh with an accuracy reaching ± 0.05 mm. Hong et al. [16] acquired the point cloud model of 3D zigzag-line welding seams based on laser displacement sensing, which had a high accuracy while the high cost of the sensor limits its application. Cameras with lower costs can be a good replacement considering the practice in the industry. Kim et al. [17] and Zhou et al. [18] obtained the point cloud of the weldment through the RGB-D camera, Chen et al. [19] established the point cloud model of straight-line weld seams with the binocular structured light sensor, and Jing et al. [20] developed a binocular camera system to obtain point clouds of weldments. In our previous work, the point cloud model of the weldment was obtained by 3D reconstruction of the weldment from monocular camera images.

The accuracy and data size of the point cloud obtained by the above methods are quite different. As a result, methods of processing the point cloud to obtain weld seam information also differ. For the point cloud acquired by a laser vision sensor, the point cloud data have a high accuracy, and the point cloud is acquired frame by frame [21]. Point cloud processing can be performed both on a few frames of data or multiple frames. For the butt weld seam, ref. [22] processed the point cloud data frame by frame after completing the initial point location of the weld and finally realized the weld tracking by extracting the intersection of each frame. This real-time processing method easily achieved feature extraction. Correspondingly, ref. [23] chose to process multi-frame data for 3D weld seam and obtained the entire weld seam information through a moving average filter and two stages of feature recognition, and finally sent the processing path to the manipulator to realize automatic processing. In contrast, point cloud obtained by RGB-D cameras and photography has lower precision and more noise, which requires more

complex algorithms to process. Ref. [18] used an RGB-D camera to reconstruct the weldment point cloud model and then proposed an edge intensity gradient-based algorithm to extract the weld seam. There are fewer related studies on photography compared with other methods. Photography is a method of reconstructing point clouds using multiple 2D images, and noise exists in its generated point cloud model. Ref. [24] dealt with the weldment point cloud obtained by the photographic method and proposed a processing method that combined 3D geometrical features and machine learning classifiers, and finally completed the weld classification. However, in this paper, only classification of the welds was carried out, and location information of the welds was not accurately extracted.

The above studies mainly deal with point cloud models of straight and spatial curve welds, but there are few studies on annular weld seams. Unlike straight welds, annular weld seams are usually composed of a large number of small circles, resulting in heavy mounts and a large range of point cloud data. At the same time, there are differences in the welding method of annular weld seams compared to straight welds and spatially curved welds. During the welding process of annular weld seams, the welding torch performs a circular motion, which results in the identification of circular welds focusing on the center position of the weld seam. For laser vision sensor circular weld point cloud data, Ma et al. [25] proposed a self-updating template matching (SUTM) method to obtain weld feature points, and a spatial circle center fitting algorithm based on random sampling consistency (RANSAC), obtaining a good effect on the weld seam point cloud with less noise, but requiring improvements for that with high noise. Similarly, Tan et al. [26] proposed a method based on the minimization of the boundary area of local scanning data points and completed the cylindrical tube detection with the reference ideal model that also required high accuracy of the point cloud. The research object of this paper is the tubesheet, as shown in Figure 1, which is composed of plenty of annular welds, characterized by a large number of welds and a large area. In our previous work [23,24], a monocular camera was used to capture the annular weld seam image, and its processing was performed to extract the weld information. However, our previous work has the following limitations: firstly, only a single weld seam can be captured and processed at a time. Secondly, the object type processed in the previous work was an image, which differs from the point cloud data in this paper.

In summary, the current point cloud annular weld detection has the following problems. First, there are few studies on annular weld detection, and second, weld seam detection in the noisy point cloud does not obtain a good effect. Furthermore, the tube sheet is not suitable for collecting point clouds with a laser vision sensor considering its character-

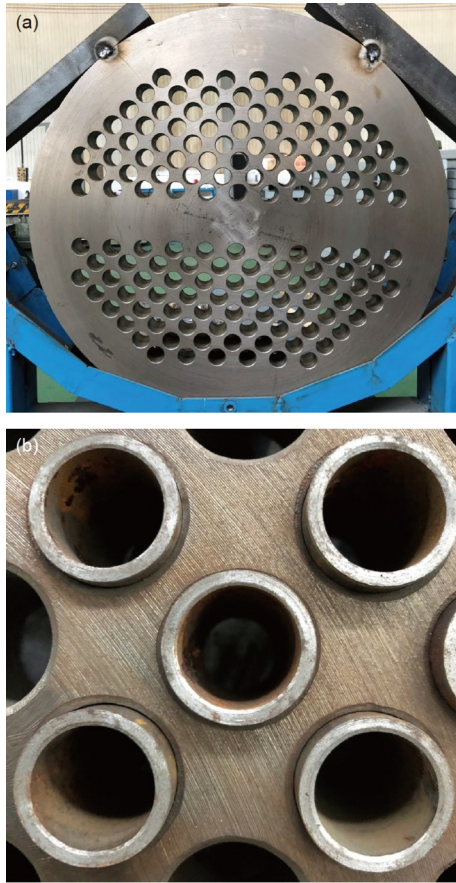


Figure 1 (Color online) Figure of tube sheet. (a) Panoramic picture of tube sheet; (b) annular weld seam in tube sheets.

istics. Relatively large noise exists in the obtained point cloud of the tubesheet by photography in our previous work. Aiming at the above problems, this paper proposed a point cloud annular weld seam detection network based on deep learning, which improved the robustness of point cloud annular weld seam detection.

2 Related work

2.1 Point cloud acquisition

Common point cloud acquisition methods include lidar [27], laser vision sensors [21], RGB-D cameras [28], and vision-based 3D reconstruction [29]. Among them, the point cloud obtained by lidar is relatively sparse and mainly used in automatic driving scenarios; the scanning range of a laser vision sensor is small, which is suitable for scenes with a small-range welding seam. Due to the characteristics of multiple annular seams in the tube sheet, it is more suitable to use a camera to reconstruct the point cloud model. There are two sources of point cloud models used in this paper, the first is obtained by 3D reconstruction with a monocular camera, and the second by shooting with an RGB-D camera. In our

previous research, by using a monocular camera and a laser rangefinder, a new method for 3D dense point cloud reconstruction was proposed, and the dense point cloud model of the weldment was finally obtained [30,31], as shown in Figure 2. In contrast, the method using RGB-D is simpler and can be directly photographed by an RGB-D camera, but the result is not stable, as shown in Figure 3. Figure 3(a) shows a more precise point cloud of the weldment obtained by using the RGB-D camera, and as a comparison, the point cloud in Figure 3(b) was much poorer.

However, the annular weld seam point cloud obtained by

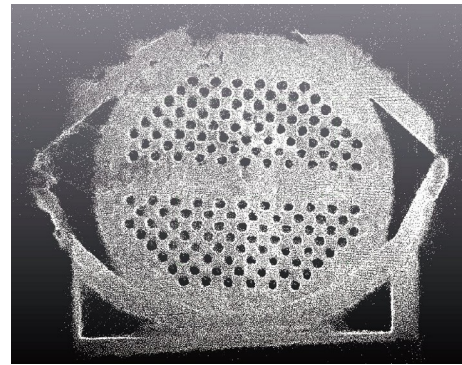


Figure 2 (Color online) Weldment point cloud obtained by 3D reconstruction of monocular camera.

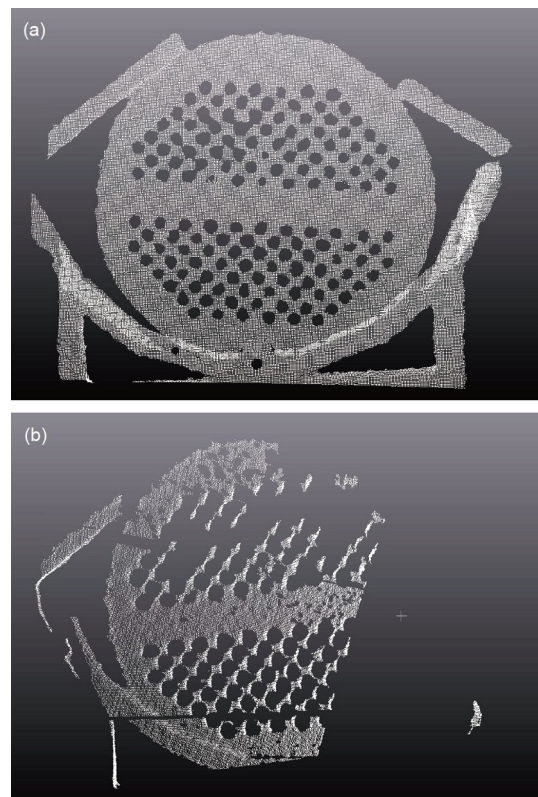


Figure 3 (Color online) Weldment point cloud obtained by RGB-D camera.

the above two methods may have the following phenomena: the first is that the annular weld seam is deformed and fails to obtain a standard circle, as shown in Figure 4(a); the second is that a large number of noise points exist in the annular weld seam, affecting the weld seam detection, as shown in Figure 4(b); the third is that some voids are in the point cloud, causing misjudgment of welds, as shown in Figure 4(c). All of these have brought obstacles to the detection of circumferential seams and have a huge impact on the success rate of weld detection.

2.2 Point cloud processing method

The traditional point cloud processing methods are mainly based on local features and global features. The basic idea is to extract features from the filtered point cloud, and then perform feature matching with the marked categories in the point cloud library. Ref. [32] proposed a feature point extraction method for complex models and constructed feature lines for segmenting point clouds. Ref. [33] defined three pattern vectors according to the normal angles of adjacent points and proposed a method to identify the curve profile on the profile. Ref. [34] applied RANSAC for circular fitting. However, considering the defects of the annular weld point cloud processed in this paper, the traditional point cloud processing method has a low success rate, so the deep learning method is considered to process the weld point cloud. Deep learning point cloud processing methods can be mainly divided into point cloud compression methods [35] and direct point cloud processing methods [36]. In the field of defect detection, ref. [35] compressed the point cloud into two-dimensional data and used the U-net network to extract

the point cloud circular defects. The direct point cloud processing methods include point-based methods and voxel-based methods. Due to the large quantity of points in the point cloud, point-based processing methods have higher computational requirements. In contrast, voxel-based methods divide the point cloud into a certain number of voxels, and after random sampling and normalization of points, use several VFEs (voxel feature encodings) for each non-empty voxel for local feature extraction, thus achieving the purpose of reducing the amount of calculation.

3 Method

In this chapter, the WeldNet network architecture proposed in this paper will be described, and the relevant training details will be introduced.

3.1 WeldNet architecture

The WeldNet architecture is mainly divided into the following parts: voxel division, VFE, sparse convolutional middle layers, and region proposal network (RPN), as shown in Figure 5 and introduced separately below.

Voxel division In this part, the point cloud space will be divided into voxels of equal size, and then the points in the point cloud will be assigned to the corresponding voxels. Assuming that the value ranges of the point cloud in the X , Y , and Z directions are D , H , and W , and the size of each voxel is set to v_D , v_H , and v_W , then the number of voxel grids is

$$\begin{aligned} D' &= \frac{D}{v_D}, \\ H' &= \frac{H}{v_H}, \\ W' &= \frac{W}{v_W}. \end{aligned} \quad (1)$$

Then, the points in the point cloud are traversed and assigned to the corresponding voxel. In this paper, v_D , v_H , and v_W are set to 1 mm, 1 mm, 1 mm, respectively. For a voxel that exceeds a certain number T , in order to reduce the computational workload and improve the running speed, random sampling is performed on this voxel, so that the number of points is reduced to T . In this paper, T is set to 20.

Voxel feature extraction The voxel feature extraction uses the VFE layer, and the VFE layer can extract the voxel features. The VFE layer first takes all the points in a voxel as input and calculates the mean of voxel midpoints as centroids, denoted as (v_x, v_y, v_z) , and then the offset of each point in the voxel relative to the centroid is calculated. In ref. [37], the input feature set was a 7-dimensional matrix, but it has been improved in this paper to remove the redundant reflectivity information and obtain a 6-dimensional input fea-

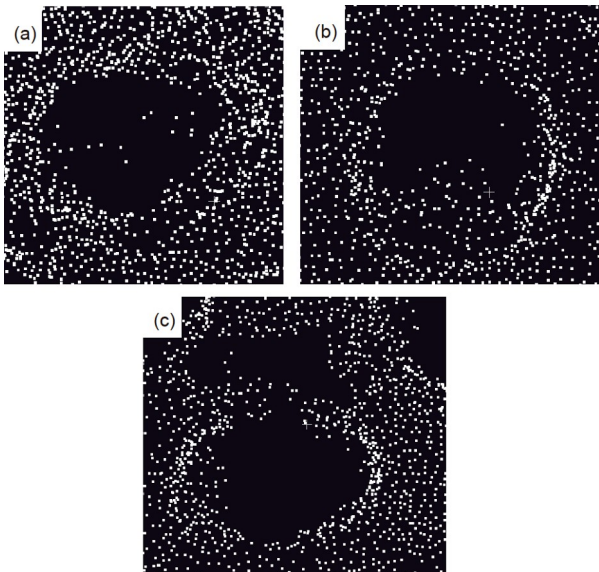


Figure 4 Defects in annular weld seam point clouds. (a) Non-standard circle; (b) multi-noise; (c) unexpected void.

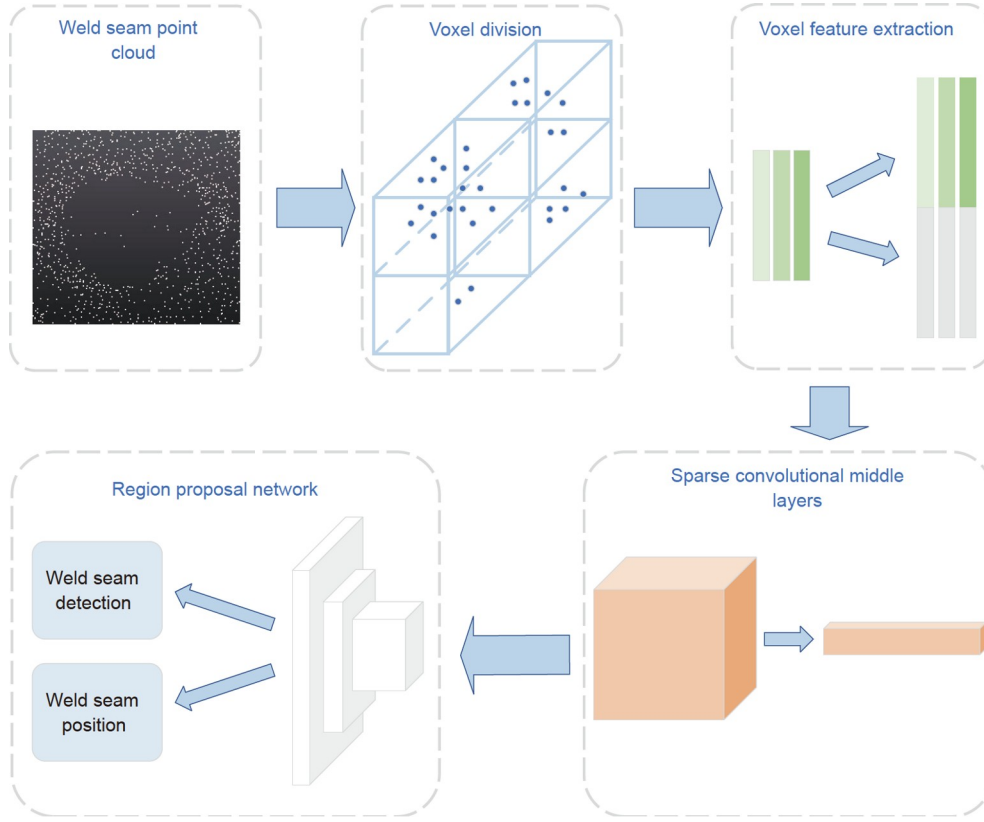


Figure 5 (Color online) The WeldNet architecture.

ture set matrix, as shown in eq. (2),

$$V_{in} = \{p_i = [x_i, y_i, z_i, x_i - v_x, y_i - v_y, z_i - v_z]^T \in \mathbb{R}^6\}_{i=1, \dots, t}, \quad (2)$$

where p_i is the point in voxel. After obtaining the input feature set, a fully connected network (FCN) is used, including a linear layer, a batch normalization (BN) layer, and a rectified linear unit (ReLU) layer, and after that, point-wise features of the voxel are extracted. Then, local aggregated features for each voxel are obtained with elementwise max pooling. Finally, the resulting local aggregated features are tiled and these tiled features are concatenated with point-wise features, and the output feature set is obtained, as shown in eq. (3),

$$\begin{aligned} f_i^{out} &= [f_i^T, f'^T]^T, \\ V_{out} &= \{f_i^{out}\}_{i=1, \dots, t}, \end{aligned} \quad (3)$$

where f_i is the point features get from the FCN, and f' is the locally aggregated features get from the elementwise max pooling. The whole VFE process is shown in Figure 6. In the figure, the green boxes represent the input features corresponding to different points.

Sparse convolutional middle layers Convolutional neural networks have been proven to be very effective for 2D

image signal processing and are used in a lot of well-established applications, but unlike images, 3D point clouds are much sparser. In order to improve computational efficiency, ref. [38] proposed sparse convolution. Sparse convolutional middle layers mainly contain two parts of sparse convolution, which convert sparse 3D information into 2D bird's eye view (BEV) images by down-sampling on the z -axis. As sparse convolution only computes non-zero elements instead of all elements, it greatly improves computational efficiency. The sparse point cloud data are transformed into dense feature maps by sparse convolution.

Region proposal network RPN has been used in many object detection tasks [39,40]. In RPN, each point in the input feature map is assigned several anchor boxes as initial detection boxes, and then the convolutional network is used to determine whether a detection box contains recognized objects, finally, the detection box is fine-tuned using bounding box regression. Based on the input of the feature map, RPN can select the area that contains the target as well as the corresponding probability, and complete the target recognition by comparing with the set threshold. The threshold is divided into a positive threshold and a negative threshold, which are used to identify the target and calculate the loss. The RPN architecture used in this paper is shown in Figure 7. In the figure, the light yellow boxes represent

$$L = \alpha \frac{1}{N_{\text{pos}}} \sum_i L_{\text{cls}}(p_i^{\text{pos}}, 1) + \beta \frac{1}{N_{\text{neg}}} \sum_j L_{\text{cls}}(p_j^{\text{neg}}, 0) + \frac{1}{N_{\text{pos}}} \sum_i L_{\text{reg}}(u_i, u_i^*), \quad (7)$$

where N_{neg} is the negative anchor, p_i^{pos} and p_j^{neg} are the softmax outputs for positive anchor and negative anchor, u_i and u_i^* are the regression output and truth for positive anchor, L_{cls} is the binary cross entropy loss, α and β are constants, L_{reg} is the regression loss, which is calculated with the SmoothL1 function in this paper. In the loss function, the first two items are the weld seam identification error, and the last item is the weld seam position error.

3.3 Data augmentation

The current public point cloud datasets are mainly autonomous driving datasets such as Kitti. There are few studies on point cloud-based weld seam identification, and the study on tubesheet annular weld seams is even rarer. Therefore, it is difficult to find a suitable dataset for this paper. However, deep learning requires a large number of data for training, and it is difficult to meet the requirements only by collecting point clouds by our team. To conquer this problem, this paper adopts a data augmentation method. For point clouds, commonly used data augmentation methods include rotation, noise addition, downsampling, etc. The method of noise addition requires setting an appropriate mean and variance, which is difficult to grasp. Therefore, the rotation method is mainly applied in this paper. Each rotation is set to 6 degrees, and a total of 60 degrees of rotation gets 10 times the data, which will meet the need in this paper.

3.4 Training details

In voxel-based point cloud recognition, different anchor sizes need to be set for different targets. For the annular weld in this paper, the anchor size is set as

$$l = 20 \text{ mm}, w = 20 \text{ mm}, h = 5 \text{ mm}. \quad (8)$$

For annular weld seams of different sizes, the anchor size needs to be modified accordingly to improve the recognition success rate. The threshold for the positive anchor of the weld seam is set to 0.6, and the threshold for the negative anchor is set to 0.4. The anchors between 0.4 and 0.6 are ignored during training, while others are used to calculate the loss with eq. (7). The training in this paper adopts the stochastic gradient descent (SGD) method, and the optimizer used is the Adam optimizer. The learning rate is 0.0005, and the model is trained for 100 epochs.

4 Experiment and discussion

In this chapter, the annular weld seam dataset is first estab-

lished, and then the dataset is used to train the network in this paper, and comparison experiments are carried out with other algorithms.

4.1 Dataset creation

The tube sheet used in this paper has 144 annular weld seams. Through the point cloud acquisition method described in Section 2.1, 4 sets of valid point cloud models are collected, corresponding to 576 annular weld seams. By the data augmentation method described in Section 3.3, the size of data is expanded by a factor of 10, that is, a total of 5760 sets of annular weld seam point cloud data. Dividing these data according to the ratio of 4:1, training dataset and test dataset are obtained respectively. The data set is composed of point cloud data and actual weld seam box. By comparing the detected weld seam with the actual weld seam, the loss can be calculated by eq. (7).

4.2 Network training

The CPU used in this paper is i7-9750, and the GPU is NVIDIA GeForce GTX 1650. The training process is shown in Figure 8. It can be seen that the initial loss of training is very large. As the training progresses, the loss continues to decrease, and the training gradually converges after 40 epochs.

After the training is completed, the model is tested with the test dataset, and some of the test results are shown in Figures 9–11, which correspond to the three defect problems proposed in Figure 4, respectively. Among them, Figure 9 shows the case of a non-standard circle in the weld point cloud, Figure 10 shows the case of multi-noise in the point cloud, and Figure 11 shows the case of unexpected voids. In these figures, the yellow box represents the detection box, and the orange circle represents the location of the weld seam. It can be seen that the proposed network obtained a good detection

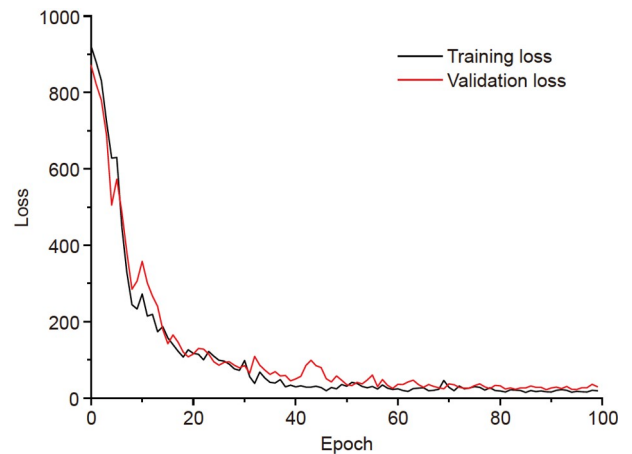


Figure 8 (Color online) Training loss of WeldNet.

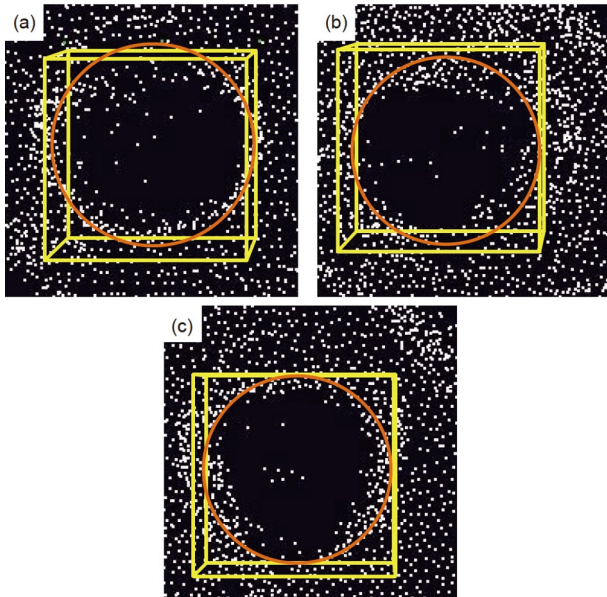


Figure 9 (Color online) WeldNet annular weld seam point cloud recognition test results in the case of non-standard circular shape.

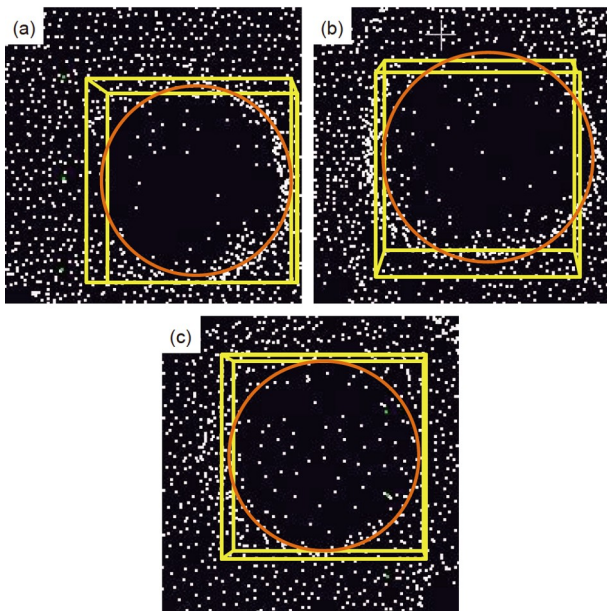


Figure 10 (Color online) WeldNet annular weld seam point cloud recognition test results in the case of multi-noise.

effect for the annular weld point cloud model, and performs well in the above three problems. In [Figure 9](#), the edge shape of the point cloud is marked by a green polyline. As shown, the shape of these point clouds is very irregular, which is obviously inconsistent with the actual weld seam position, but WeldNet has completed the recognition. In [Figure 11](#), the positions of the voids in the point cloud are marked by green ellipses. These holes are distributed around the weld and have an impact on the weld seam detection. Similarly, WeldNet has completed the weld seam detection with a good effect. On the test dataset, WeldNet achieved a detection

accuracy of 90.6%.

4.3 Comparison with RANSAC point cloud processing methods

The RANSAC algorithm has a lot of applications in point cloud processing and can be used to detect the basic shape in the point cloud [\[39\]](#). In this experiment, the RANSAC algorithm was used to extract the annular weld seam in the point cloud, and some of the results are shown in [Figure 12](#). The original point cloud is shown in white, the circular point detected by RANSAC is shown in yellow, and the orange

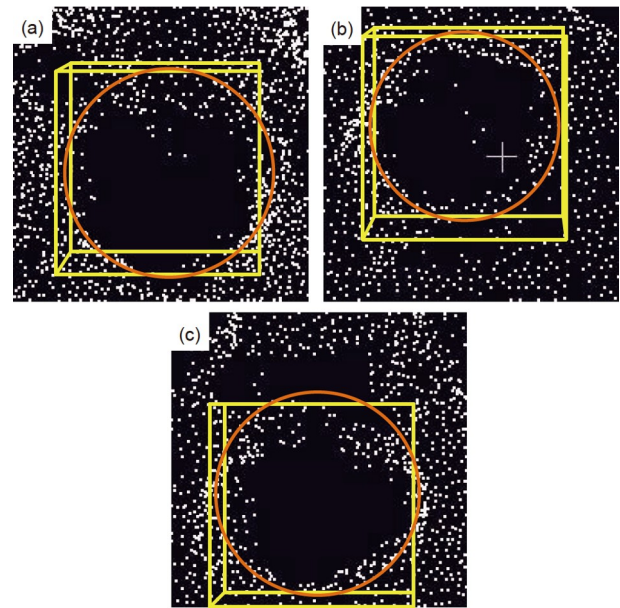


Figure 11 (Color online) WeldNet annular weld seam point cloud recognition test results in the case of voids.

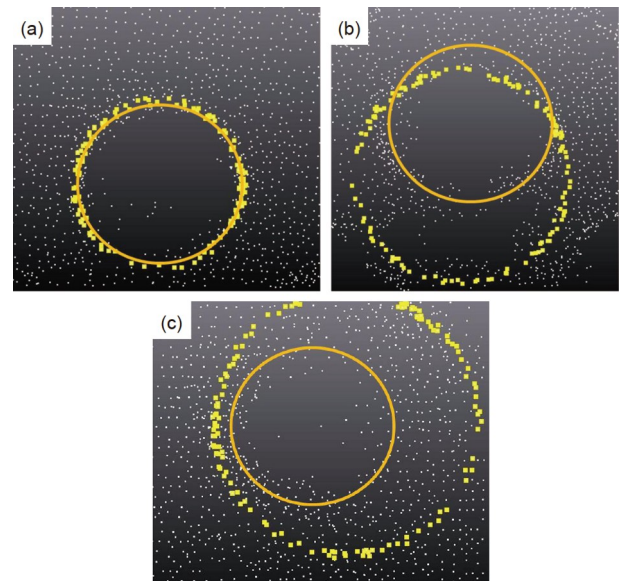


Figure 12 (Color online) RANSAC weld point cloud recognition results.

circle is the theoretical weld seam position. It can be seen from the figure that, compared with WeldNet in Figures 9–11, the detection result of the RANSAC algorithm is more unstable. In Figure 12(a), there are fewer noise points, and RANSAC extracts the circles well. In Figure 12(b), the point cloud is similar to that in Figure 11, an error void is abnormally under the weld point cloud. In this case, the RANSAC algorithm fails to detect the correct weld seam position. In Figure 12(c), due to the existence of numerous noise points, the RANSAC algorithm also fails to extract the accurate weld seam position.

For RANSAC and WeldNet, the errors of weld identification are compared, as shown in Table 1. It can be clearly seen in the table that compared with the traditional RANSAC algorithm, WeldNet has better robustness and higher welding seam recognition accuracy. Due to the large number of noise points and irregular voids that cause significant interference to the RANSAC method, the RANSAC algorithm is unable to obtain the optimal solution. In contrast, WeldNet has a stronger anti-interference ability against noisy points due to its analysis and processing of point clouds in voxels, achieving better robustness. In this paper, the welding seam identification error exceeding 10 mm is regarded as the identification failure, and the identification success rate of the two algorithms is obtained. It can be seen that compared with RANSAC, the success rate of network identification proposed in this paper has increased by 23%.

4.4 Comparison with U-Net in ref. [35]

In ref. [35], the author applied U-net to process the point cloud, which can realize point cloud circular defect detection. In this experiment, the point cloud detection effects of U-net and Weldnet were compared, as shown in Figure 13. The original point cloud in Figure 13 is the defects mentioned in Figure 4, respectively. The second column in Figure 13 is the detection result of the original point cloud with the U-Net network, and the third column is the detection result with the WeldNet, where the orange circle is the actual position of the weld seam. It can be seen that the detection results of U-net are not good in various defect situations, the detection is not clear in the multi-noise situation, and mis-detection occurs in the multi-void situation. U-Net performs well in processing relatively regular point clouds, but its recognized performance is not good for the tubesheet point cloud model in this paper under the influence of multiple

defect data. An important indicator for evaluating U-Net is IOU (Intersection over Union). In order to better evaluate U-Net and WeldNet, the IOU of the two methods was compared, as shown in Table 2. It can be seen that the IOU of U-Net is 65.7%, and the IOU of WeldNet is 83.5%, which is significantly improved.

4.5 Comparison with advanced point cloud target recognition networks

The current advanced point cloud target recognition networks include VoxelNet [37], PointNet++ [40], and PV-RCNN [41]. The comparison between the above network and WeldNet is shown in Table 3. Among them, PointNet is a point-based processing method mainly used for identifying vehicles and pedestrians in LiDAR point clouds, which has not yet been applied in weld seam recognition and cannot

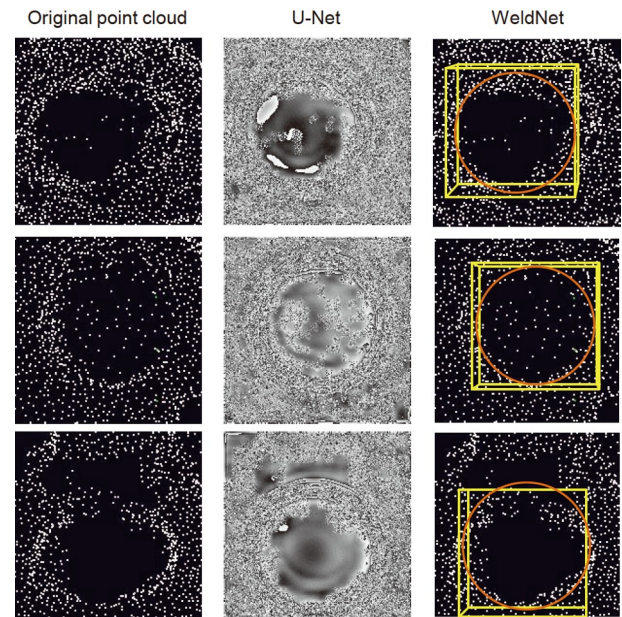


Figure 13 (Color online) Comparison of this paper with ref. [35].

Table 2 IOU comparison between U-Net and WeldNet

Methods	IOU (%)
U-Net	65.7
WeldNet	83.5

Table 3 Comparison between advanced networks and WeldNet

Methods	Time (s)	IOU (%)
PointNet++	N/A	N/A
VoxelNet	0.56	68.5
PV-RCNN	0.23	85.6
WeldNet	0.13	83.5

Table 1 Error comparison between RANSAC and WeldNet

Method	Mean error (mm)	Standard deviation of error (mm)	Identification rate (mm)
RANSAC	5.7	4.56	67.6
WeldNet	2.2	1.12	90.6

complete circular weld seam recognition. As for VoxelNet and PV-RCNN, the identification of annular weld seams is able to be completed, and the recognition results are shown in Table 3. It can be seen that VoxelNet has a slower computational speed due to its use of 3D convolution, and its recognition performance needs to be improved. PV-RCNN achieved good recognition performance due to its two-step strategy, but its computational time is significantly longer compared to WeldNet. Overall, the above experiments show that WeldNet achieves good recognition results at a fast speed.

5 Conclusions

Groove textures with different area densities were fabricated on steel samples. Friction tests were conducted under dry sliding conditions, and temperature rise, friction coefficient and wear of both the textured and untextured specimens were measured. Compared to the untextured specimen, the temperature rise observed in the textured specimens was obviously smaller. The specimen with high texture density exhibited a low temperature rise. The difference in temperature rise between the textured and untextured samples was attributed to two main reasons: changes in heat dissipation and the allocation of energy between the tribo-pairs due to the textured structures. The energy consumed by wear and plastic deformation was small compared to the total energy input by friction, thus, their effect on temperature rise can be considered to be negligible.

The development of society urgently requires the demand for welding automation, and the traditional “teach-playback” robot fails to meet the actual needs. A lot of research on automatic weld seam inspection has been studied, however, few studies on weld seam inspection of tube sheet annular weld seams have been conducted. Due to its characteristics, the point cloud model established for the annular weld seam of the tube sheet has many defects, bringing obstacles to the weld seam inspection.

The purpose of this paper is to solve the problem of welding seam detection in the point cloud model of tube sheet annular weld seam. To this end, this paper proposed an annular weld seam point cloud detection network WeldNet. WeldNet consists of four parts, voxel division, voxel feature extraction, sparse convolutional middle layers, and region proposal network. This paper proposed a new VFE layer, adopted a sparse convolutional layer to improve the computational speed and an RPN network to detect the weld seam, and designed a loss function for annular weld seams. Regarding the lack of a weld seam dataset, this paper established an annular weld seam dataset and trained WeldNet on this basis. The test result showed that the proposed model can be a good solution to the cases of the three defects in the

point cloud. Compared with the traditional RANSAC method, WeldNet has obtained improvements in both weld seam identification error and the weld seam identification success rate. Among them, the weld seam identification success rate has increased by 23%. Compared with U-Net of ref. [33], WeldNet has a better detection effect, and the weld seam detection IOU is improved by 17.8%. Compared with advanced point cloud target recognition networks, the calculation speed is greatly improved while achieving similar results.

This paper provides a new solution for point cloud detection of annular weld seams and achieves better results. This research contributes to improving the detection level of annular weld seams and can shed light on the practice of welding automation. However, the types of weld seams currently supported by the network are limited to the annular weld seam, and other weld types require future studies. At the same time, in future work, the detection success rate of annular weld seams also needs to be further improved.

This work was supported by the Key Research & Development Plan of China (Grant No. 2022YFB3404800), the Key Research & Development Plan of Hubei Province (Grant No. 2021BAA195), and the National Natural Science Foundation of China (Grant No. 52188102).

- 1 Wang B, Hu S J, Sun L, et al. Intelligent welding system technologies: State-of-the-art review and perspectives. *J Manuf Syst*, 2020, 56: 373–391
- 2 Yang L, Liu Y, Peng J. Advances techniques of the structured light sensing in intelligent welding robots: A review. *Int J Adv Manuf Technol*, 2020, 110: 1027–1046
- 3 Li J, Jin S, Wang C, et al. Weld line recognition and path planning with spherical tank inspection robots. *J Field Robotics*, 2022, 39: 131–152
- 4 Vithanage R K W, Mohseni E, Qiu Z, et al. A phased array ultrasound roller probe for automated in-process/interpass inspection of multipass welds. *IEEE Trans Ind Electron*, 2021, 68: 12781–12790
- 5 Yang L, Liu Y, Peng J, et al. A novel system for off-line 3D seam extraction and path planning based on point cloud segmentation for arc welding robot. *Robotics Comput-Integrated Manuf*, 2020, 64: 101929
- 6 Fan J, Deng S, Ma Y, et al. Seam feature point acquisition based on efficient convolution operator and particle filter in GMAW. *IEEE Trans Ind Inf*, 2021, 17: 1220–1230
- 7 Fan J, Deng S, Jing F, et al. An initial point alignment and seam-tracking system for narrow weld. *IEEE Trans Ind Inf*, 2020, 16: 877–886
- 8 Ma Y, Fan J, Yang H, et al. An efficient and robust complex weld seam feature point extraction method for seam tracking and posture adjustment. *IEEE Trans Ind Inf*, 2023, 19: 10704–10715
- 9 Ma Y, Fan J, Zhou Z, et al. WeldNet: A deep learning based method for weld seam type identification and initial point guidance. *Expert Syst Appl*, 2024, 238: 121700
- 10 Wen W W, Zhang G, Hsu L T. GNSS NLOS exclusion based on dynamic object detection using LiDAR point cloud. *IEEE Trans Intell Transp Syst*, 2021, 22: 853–862
- 11 Lin G, Tang Y, Zou X, et al. Color-, depth-, and shape-based 3D fruit detection. *Precision Agric*, 2020, 21: 1–17
- 12 Štular B, Eichert S, Lozić E. Airborne LiDAR point cloud processing for archaeology. pipeline and QGIS toolbox. *Remote Sens*, 2021, 13:

- 3225
- 13 Zanjani F G, Pourtaherian A, Zinger S, et al. Mask-MCNet: Tooth instance segmentation in 3D point clouds of intra-oral scans. *Neuro-computing*, 2021, 453: 286–298
 - 14 Gao J, Li F, Zhang C, et al. A method of D-type weld seam extraction based on point clouds. *IEEE Access*, 2021, 9: 65401–65410
 - 15 Geng Y, Zhang Y, Tian X, et al. A method of welding path planning of steel mesh based on point cloud for welding robot. *Int J Adv Manuf Technol*, 2021, 116: 2943–2957
 - 16 Hong B, Jia A, Hong Y, et al. Online extraction of pose information of 3D zigzag-line welding seams for welding seam tracking. *Sensors*, 2021, 21: 375
 - 17 Kim J, Lee J, Chung M, et al. Multiple weld seam extraction from RGB-depth images for automatic robotic welding via point cloud registration. *Multimed Tools Appl*, 2021, 80: 9703–9719
 - 18 Zhou P, Peng R, Xu M, et al. Path planning with automatic seam extraction over point cloud models for robotic arc welding. *IEEE Robot Autom Lett*, 2021, 6: 5002–5009
 - 19 Chen L, Li X, Zhou X. A 3D seam extraction and tracking method based on binocular structured light sensor. In: Twelfth International Conference on Digital Image Processing (ICDIP 2020). Osaka, 2020. 72
 - 20 Ou J, Zou L, Wan Q H, et al. Weld-seam identification and model reconstruction of remanufacturing blade based on three-dimensional vision. *Adv Eng Inf*, 2021, 49: 101300
 - 21 Liu C, Wang H, Huang Y, et al. Welding seam recognition and tracking for a novel mobile welding robot based on multi-layer sensing strategy. *Meas Sci Technol*, 2022, 33: 055109
 - 22 Zou Y, Chen J, Wei X. Research on a real-time pose estimation method for a seam tracking system. *Optics Lasers Eng*, 2020, 127: 105947
 - 23 Feng H, Ren X, Li L, et al. A novel feature-guided trajectory generation method based on point cloud for robotic grinding of freeform welds. *Int J Adv Manuf Technol*, 2021, 115: 1763–1781
 - 24 Rodriguez-Gonzalez P, Rodriguez-Martin M. Weld bead detection based on 3D geometric features and machine learning approaches. *IEEE Access*, 2019, 7: 14714–14727
 - 25 Ma Y, Fan J, Yang H, et al. A fast and robust seam tracking method for spatial circular weld based on laser visual sensor. *IEEE Trans Instrum Meas*, 2021, 70: 1–11
 - 26 Tan Y Z, Pang C K, Al Mamun A, et al. Precise pose and assembly detection of generic tubular joints based on partial scan data. *Neural Comput Applic*, 2022, 34: 5201–5211
 - 27 Wen C, Li X, Yao X, et al. Airborne LiDAR point cloud classification with global-local graph attention convolution neural network. *ISPRS J Photogrammetry Remote Sens*, 2021, 173: 181–194
 - 28 Teng X, Zhou G, Wu Y, et al. Three-dimensional reconstruction method of rapeseed plants in the whole growth period using RGB-D camera. *Sensors*, 2021, 21: 4628
 - 29 Bao Y, Lin P, Li Y, et al. Parallel structure from motion for sparse point cloud generation in large-scale scenes. *Sensors*, 2021, 21: 3939
 - 30 Wang H, Lei T, Rong Y, et al. Arc length stable method of GTAW based on adaptive Kalman filter. *J Manuf Processes*, 2021, 63: 130–138
 - 31 Wang H, Xu J, Huang Y, et al. Multilayer positioning strategy for tubesheet welding robot based on point cloud model. *IEEE Sens J*, 2023, 23: 13728–13737
 - 32 Ji X, Zhang X, Hu H. Point cloud segmentation for complex micro-surfaces based on feature line fitting. *Multimed Tools Appl*, 2021, 80: 4553–4578
 - 33 Fan J, Ma L, Sun A, et al. An approach for extracting curve profiles based on scanned point cloud. *Measurement*, 2020, 149: 107023
 - 34 Kuželka K, Slavík M, Surový P. Very high density point clouds from UAV laser scanning for automatic tree stem detection and direct diameter measurement. *Remote Sens*, 2020, 12: 1236
 - 35 Lafiosca P, Fan I S, Avdelidis N P. Automatic segmentation of aircraft dents in point clouds. SAE Paper 2022-01-0022, 2022
 - 36 Xu Y, Tong X, Stilla U. Voxel-based representation of 3D point clouds: Methods, applications, and its potential use in the construction industry. *Automation Construction*, 2021, 126: 103675
 - 37 Zhou Y, Tuzel O. VoxelNet: End-to-end learning for point cloud based 3D object detection. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). Salt Lake City, 2018. 4490–4499
 - 38 Graham B. Submanifold Sparse Convolutional Networks. arXiv: 1706.01307
 - 39 Schnabel R, Wahl R, Klein R. Efficient RANSAC for point-cloud shape detection. *Comput Graphics Forum*, 2007, 26: 214–226
 - 40 Qi C R, Yi L, Su H, et al. PointNet plus plus: Deep hierarchical feature learning on point sets in a metric space. In: Proceedings of the 31st International Conference on Neural Information Processing Systems. Long Beach: Curran Associates Inc., 2017
 - 41 Shi S, Guo C, Jiang L, et al. PV-RCNN: Point-voxel feature set abstraction for 3D object detection. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition. Seattle: IEEE, 2020. 10526–10535